

Problem 1

Let $N(y)$ be the number events from $[0, y]$ thus $N(y) \sim \text{Poisson}(\lambda y)$

$$\begin{aligned} F_Y(y) &= \mathbb{P}(Y \leq y) = \mathbb{P}(N(y) \geq n) = 1 - \mathbb{P}(N(y) < n) \\ &= 1 - \sum_{k=0}^{n-1} \frac{e^{-\lambda y} (\lambda y)^k}{k!} \end{aligned}$$

$$\begin{aligned} f_Y(y) &= - \sum_{k=0}^{n-1} \frac{d}{dy} \frac{e^{-\lambda y} (\lambda y)^k}{k!} \\ &= -\lambda e^{-\lambda y} \left(\sum_{k=0}^{n-1} \frac{(\lambda y)^{k-1} (k - \lambda y)}{k!} \right) \\ &= -\lambda e^{-\lambda y} \left(\sum_{k=1}^{n-1} \frac{(\lambda y)^{k-1}}{(k-1)!} - \sum_{k=0}^{n-1} \frac{(\lambda y)^k}{k!} \right) \\ &= -\lambda e^{-\lambda y} \left(\sum_{m=0}^{n-2} \frac{(\lambda y)^m}{m!} - \sum_{k=0}^{n-1} \frac{(\lambda y)^k}{k!} \right) \\ &= -\lambda e^{-\lambda y} \left(\frac{-(\lambda y)^{n-1}}{(n-1)!} \right) \\ &= \frac{\lambda^n y^{n-1} e^{-\lambda y}}{(n-1)!} \end{aligned}$$

Problem 2

1. Let X be the number of earthquakes in a year from time 0 to the end of the year so thus is a Poisson distribution.

$$\begin{aligned}
 G(s) &= \sum_{k=0}^{\infty} \frac{e^{-\lambda} (s\lambda)^k}{k!} \\
 &= e^{-\lambda} \sum_{k=0}^{\infty} \frac{(s\lambda)^k}{k!} \\
 &= e^{-\lambda} e^{\lambda s} = e^{\lambda(s-1)}
 \end{aligned}$$

$$\begin{aligned}
 G(s) &= \sum_{k \text{ even}} \mathbb{P}(X = k) s^k + \sum_{k \text{ odd}} \mathbb{P}(X = k) s^k = G_{\text{even}}(s) + G_{\text{odd}}(s) \\
 G(1) &= G_{\text{even}}(1) + G_{\text{odd}}(1) = 1
 \end{aligned}$$

$$\begin{aligned}
 G(-1) &= G_{\text{even}}(-1) + G_{\text{odd}}(-1) \\
 e^{-2\lambda} &= \sum_{k \text{ even}} \mathbb{P}(X = k) (-1)^k + \sum_{k \text{ odd}} \mathbb{P}(X = k) (-1)^k = G_{\text{even}}(1) - G_{\text{odd}}(1)
 \end{aligned}$$

$$\begin{aligned}
 2G_{\text{even}}(1) &= 1 + e^{-2\lambda} \\
 G_{\text{even}}(1) &= \frac{1 + e^{-12}}{2}
 \end{aligned}$$

$$\begin{aligned}
 2. \quad \mathbb{P}(\text{no more than one earthquake every month}) &= \mathbb{P}(T_1 \leq 1) \mathbb{P}(T_2 \leq 1) \cdots \mathbb{P}(T_{12} \leq 1) \\
 &= (\mathbb{P}(T_1 \leq 1))^{12} = (e^{-\lambda/12} (\lambda/12)^0 + e^{-\lambda/12} (\lambda/12)^1)^{12} = (e^{-\lambda/12} (1 + \lambda/12))^{12}
 \end{aligned}$$

$$3. \quad \mathbb{P}(\text{A month has } k \text{ earthquakes}) = e^{-\lambda/12} \frac{(\lambda/12)^k}{k!}$$

$$\mathbb{P}(\text{Any month has } k \text{ earthquakes and others have non}) = 12 e^{-6/12} \frac{(6/12)^k}{k!} (e^{-6/12})^{11}$$

$$\begin{aligned}
 \mathbb{P}(\text{There are all earthquakes in one month}) &= \sum_{k=1}^{\infty} 12 (1/2)^k e^{-1/2} (1/k!) (e^{-1/2})^{11} \\
 &= 12 e^{-6} (e^{1/2} - 1)
 \end{aligned}$$

Problem 3

Let $X = \max(T_s, T_b)$ s.t. T_s is the time to see the first starling, and T_b is the time to see the first blackbird. Thus since this is Poisson Process, $T_b, T_s \sim$ Exponential distribution.

$$\mathbb{P}(X \leq x) = \mathbb{P}(\max(T_s, T_b) \leq x) = \mathbb{P}(T_s \leq x \text{ and } T_b \leq x) = \mathbb{P}(T_s \leq x)\mathbb{P}(T_b \leq x) = (1 - e^{-5x})(1 - e^{-2x}) = 1 - e^{-2x} - e^{-5x} + e^{-7x}$$

$$f_X(x) = 2e^{-2x} + 5e^{-5x} - 7e^{-7x}$$

$$E[X] = \int_{-\infty}^{\infty} xf_X(x) = \int_0^{\infty} x2e^{-2x} + x5e^{-5x} - x7e^{-7x} dx$$

$$= \lim_{a \rightarrow \infty} [-xe^{-2x} - (1/2)e^{-2x} - xe^{-5x} - (1/5)e^{-5x} + xe^{-7x} + (1/7)e^{-7x}]_0^a = (1/2) + (1/5) - (1/7) = (39/70)$$

Problem 4

$$1. \mathbb{P}(Z_1 = k_1, \dots, Z_n = k_n \mid X_{[0,1]} = p)$$

$\mathbb{P}(Z_1 = k_1, \dots, Z_n = k_n \mid X_{[0,1]} = p) = (\mathbb{P}(Z_1 = k_1, \dots, Z_n = k_n, X = p)) (1/\mathbb{P}(X = p))$ since we already that split of intervals of X sum up to a total number of p events the last $X = p$ is redundant, and we can get rid of it.

Thus, $= (\mathbb{P}(Z_1 = k_1, \dots, Z_n = k_n)) (1/\mathbb{P}(X = p)) = (1/\mathbb{P}(X = p)) \prod_{i=1}^n \mathbb{P}(Z_i = k_i)$ we can do this since the intervals are disjoint

$$= \frac{p!}{\lambda^p e^{-\lambda p}} \prod_{i=1}^n \frac{\lambda^{k_i} t_i^{k_i} e^{-\lambda t_i}}{k_i!} = p! \prod_{i=1}^n \frac{t_i^{k_i}}{k_i!}$$

2. $\mathbb{P}(Y_1 = k_1, \dots, Y_n = k_n)$ first number of ways to order the p variables into n boxes of size $k_1, \dots, k_n$ is $\frac{p!}{k_1! k_2! \dots k_n!}$ and then the probability of being in interval i is exactly t_i since each U_i is uniformly distributed. Since all the variables are independent we can just write them as products of all their probabilities thus

$$\mathbb{P}(Y_1 = k_1, \dots, Y_n = k_n) = \frac{p!}{k_1! k_2! \dots k_n!} (t_1)^{k_1} \dots (t_n)^{k_n} \text{ and this is clearly equal to the first one.}$$